

Polynomial Functions Quiz

Algebra 2 Version A

Name: _____

1. (4 pts) Factor completely: $x^2 - 3x + 2$

Answer: $(x - 2)(x - 1)$

2. (3 pts) Find all real zeros of $f(x) = x^2 - 3x + 2$.

Answer: 1, 2

3. (8 pts) Let $f(x) = x^3 - 2x^2 - 11x + 12$. Use algebra to answer the following.

a. Find all real zeros of f .

$-3, 1, 4$

b. Find the y -intercept.

$(0, 12)$

c. Describe the end behavior of f .

$f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, $f(x) \rightarrow$
 $+\infty$ as $x \rightarrow +\infty$

d. State the interval(s) where $f(x) > 0$.

$(-3, 1) \cup (4, +\infty)$